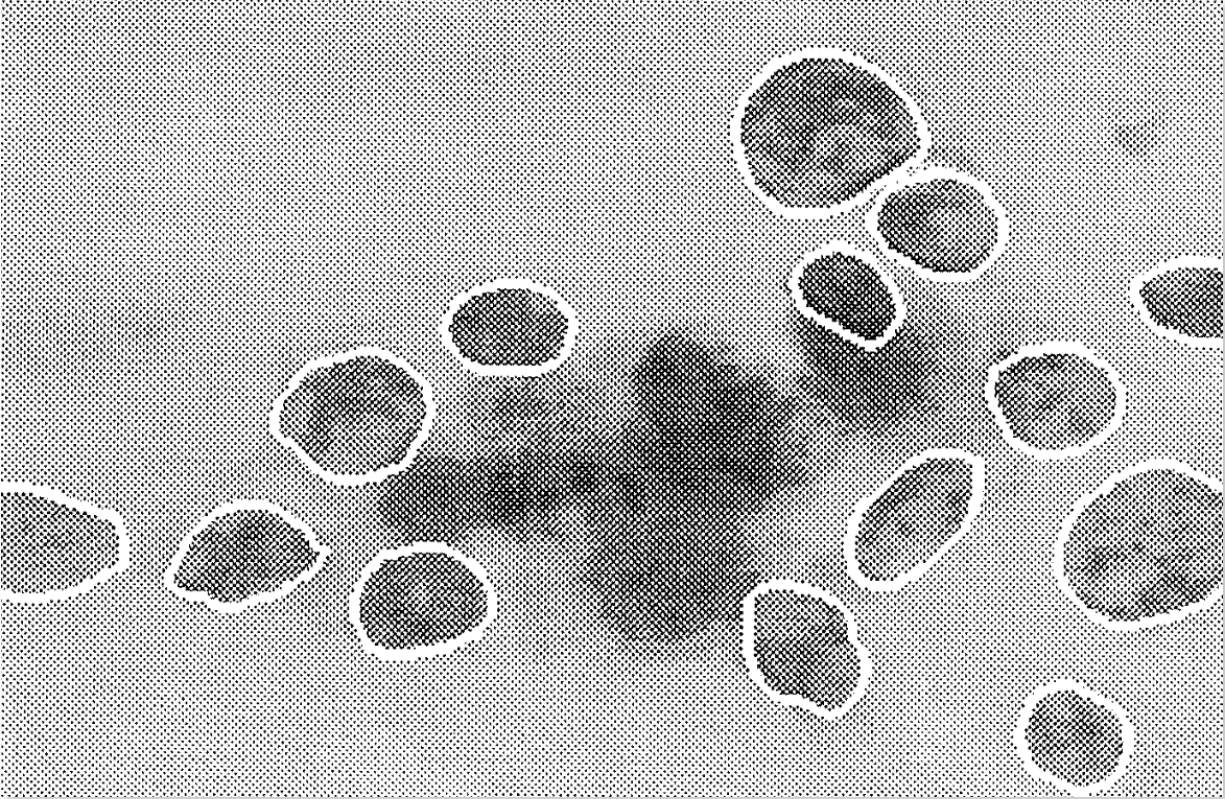


Capstone Final Report

Updated ML Model for Breast Cancer Tumor Diagnosis



Tumor cell nucleus boundaries from the report by Street, Wolberg, & Mangasarian (1992).

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Background

At the time of the report by Street, Wolberg, and Mangasarian (1992), breast masses were traditionally diagnosed by full biopsy, an invasive surgical procedure. Fine needle aspiration (FNA) allows doctors to take a much smaller tissue sample, but diagnosis based on an FNA sample was difficult because of the high number of nuclear features thought to be correlated with a malignant tumor. Street et. Al chose 3 features from the 10 obtained through FNA and developed a multi-surface method tree (MSM-Tree) model to classify the masses as malignant or benign. They achieved 97.3% accuracy, 90% sensitivity, and 86% specificity¹. Although this model's performance was strong, there is value in improving the sensitivity and specificity given the life-or-death, emotional, and financial implications of the results. With advent of random forest models, which were first described in 1995², there is potential to correctly classify tumors with even higher accuracy, sensitivity, and specificity.

Dataset Profile

This report uses the Diagnostic Breast Cancer Wisconsin Database³, hereafter called “the dataset.” The dataset contains 569 instances, each representing one patient and one breast mass. Each instance is accompanied by 30 features that were computed from a digitized image of a FNA of a breast mass. These features describe the cell nuclei in the image. There are multiple cell nuclei in each image, so the 30 features can be split into 3 categories: mean, worst, and standard error of the measurements recorded. Within each category are 10 nuclear features: 1) radius, 2) perimeter, 3) area, 4) compactness, 5) smoothness, 6) concavity, 7) concave points, 8) symmetry, 9) fractal dimension, 10) texture. The target feature was the diagnosis: benign or malignant, with 357 and 212 instances, respectively.

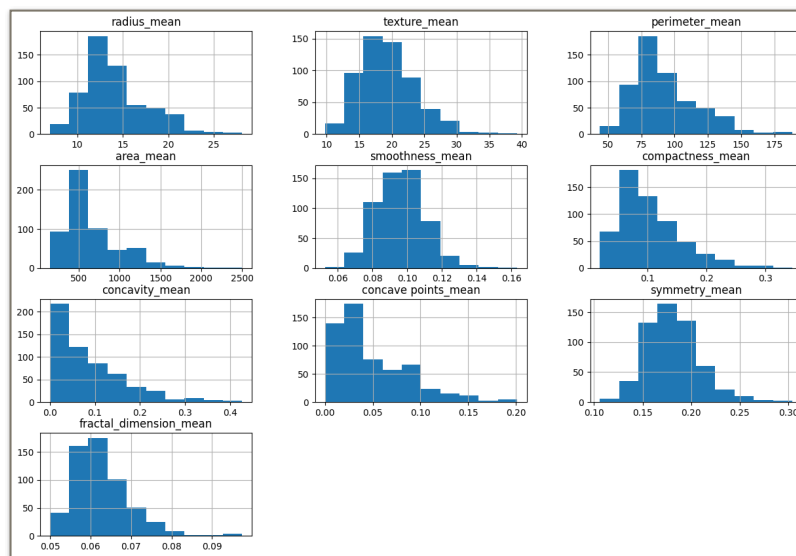
The downloadable dataset had already been cleaned. There were no duplicate instances, missing values, or impossible values.

Exploratory Data Analysis

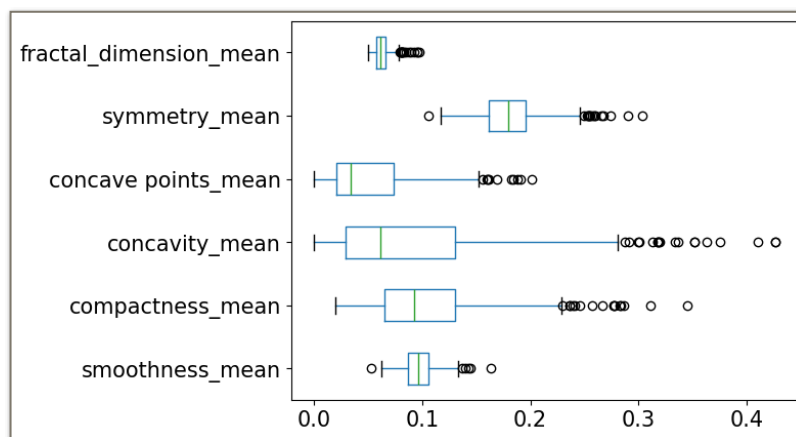
The EDA was organized by the 3 feature categories: mean, worst, and standard error.

Mean Features

All 10 mean features were right skewed to some degree, with some features showing more skewness than others. Boxplots showed some upper outliers for every feature, which made them somewhat less remarkable. Two features, symmetry and smoothness, had a single lower outlier. Perhaps more interesting were the large differences in values between features, with several features all measured below 1 and others surpassing 10, 100, and even 1000. For some models, it would be necessary to use a Scaler during feature engineering, but since the final model was a random forest, no scaling was needed.



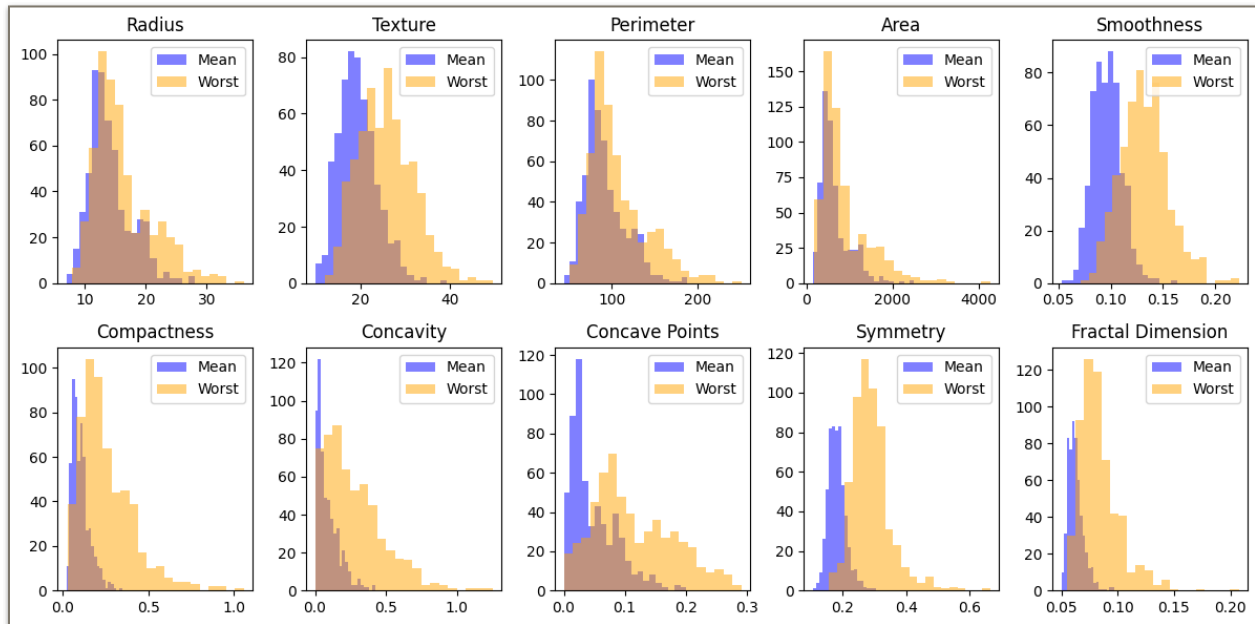
Histograms for all 10 mean features showing some level of right skewness in all features.



Boxplots for the smallest value features, showing upper outliers for all and single lower outliers for symmetry and smoothness.

Worst Features

It wasn't immediately clear if "worst" meant "larger" for each feature. To test this, worst features were overlaid onto the mean features for comparison. The result showed that worst feature values were consistently larger than mean feature values, with the smallest differences found in radius, perimeter, and area, and more notable differences for the other 7 features.



Worst feature histograms overlaid onto mean feature histograms. Worst features represent the largest measurements taken.

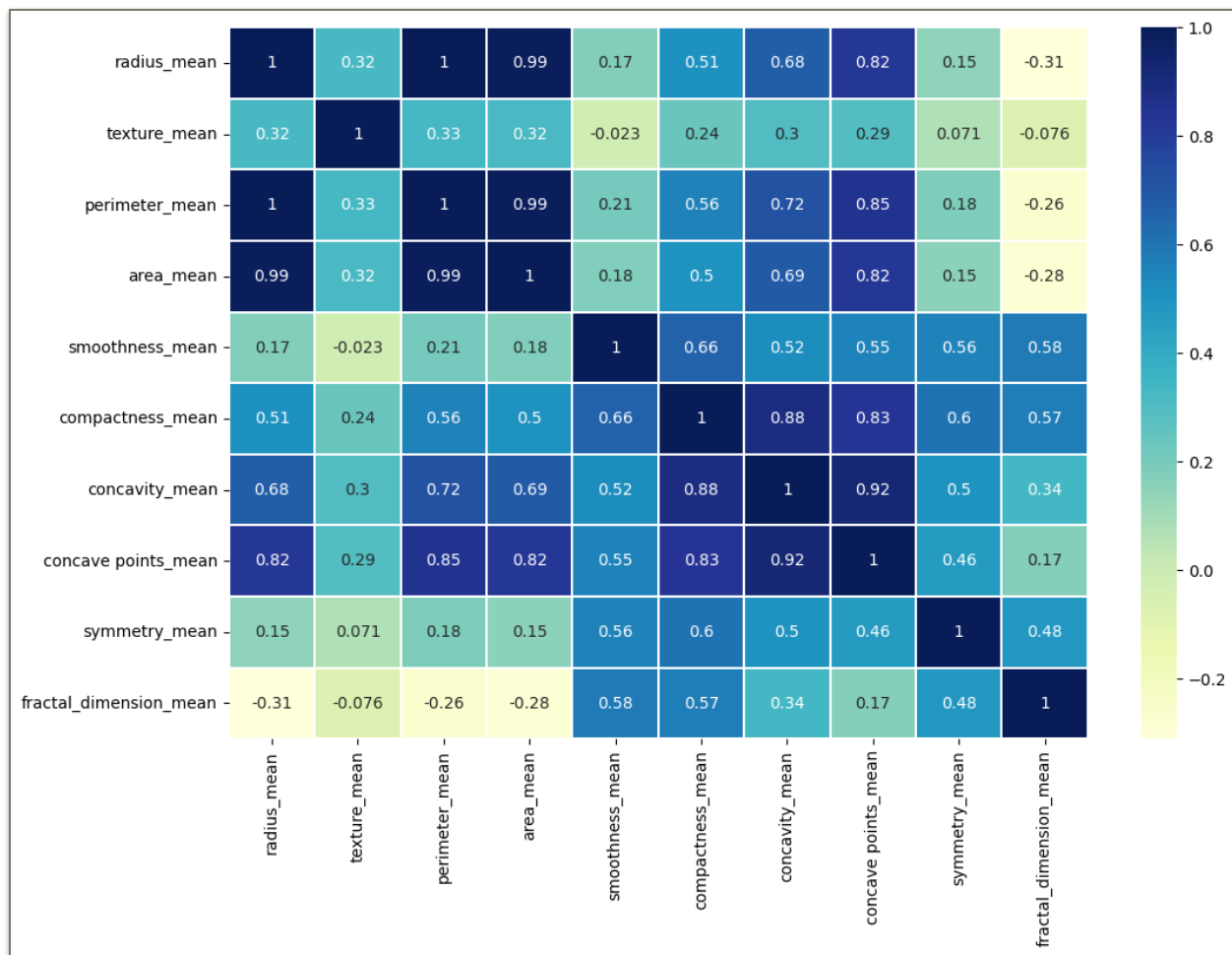
Relationships

Standard error features followed a similar pattern as the mean features, so I moved to study relationships between the features.

It was no surprise that radius, perimeter, and area were closely correlated, since they are all related geometrical equations. The concave points feature was also strongly correlated with these features, and concavity was associated with concave points. These 2 features were strongly correlated with compactness. Given such strong correlations, I noted that the final model should not include all of these features, as it would likely introduce bias and noise.

Smoothness was moderately correlated with concavity and slightly more correlated with compactness, symmetry, and fractal dimension. These 3 features were all moderately correlated with each other.

Texture was the only feature without a moderate or strong correlation. There was possibly a weak correlation with the geometric variables.



Correlation heat map of all 10 mean features.

It was also important to determine how correlated features were across linked features in the 3 categories. I.e., how closely linked were `perimeter_mean`, `perimeter_se`, and `perimeter_worst`? In the output to the right, there was a strong correlation between the linked variables. Fractal dimension and symmetry had the lowest correlations with their linked features, but still the correlations were still strong (0.77 and 0.70, respectively). Like the heat map above, this raised the question of which features to use during modeling. Such strong correlations between the 3 categories would surely add bias the model if all 30 features were included.

Correlation	
Variable	
perimeter	0.970387
radius	0.969539
area	0.959213
texture	0.912045
concave points	0.910155
concavity	0.884103
compactness	0.865809
smoothness	0.805324
fractal_dimension	0.767297
symmetry	0.699826

Feature Selection

The main challenge in building this modeling was determining which of the 30 features to use.

Attempt 1

Attempt 1 started with these guiding principles:

1. Because area, perimeter, and radius were all highly correlated with each other, I needed to select of these to represent the size of the cell nuclei. Attempt 1 used radius because it had the lowest correlation with the other 7 features.
2. Similarly, the concave points feature was strongly correlated with the 3 size variables, so it was excluded. The concavity feature was used instead, since it was only moderately correlated with radius ($R^2 = 0.68$ for concavity compared to 0.82 for concave points).

This attempt looped through different combinations of the included features in all 3 categories to test the strength of a random forest model with each feature combination while avoiding linked variables such as `radius_mean` with `radius_se`.

Unfortunately, this was an incredibly inefficient approach, so the loop was cancelled before it could complete.

Attempt 2

To improve efficiency, Attempt 2 used a similar approach to Attempt 1 with one key difference. It only looped through 7 random feature combinations. If the model accuracy didn't vary significantly between the random combinations, I reasoned I wouldn't need to test every possible combination. The better combination achieved in this attempt was `compactness_se`, `concavity_mean`, `fractal_dimension_se`, `radius_mean`, `smoothness_worst`, `symmetry_worst`, and `texture_mean` with an accuracy of 98.25%.

This attempt was faster, but there was some variation in the accuracy scores, so randomly picking a small number of combinations could be missing out on the best model. While this accuracy score is high, the goal is to build a model that is decisively stronger than the MSM Tree in the original report from Street et. Al.

Attempt 3

The original used only used 3 variables in their final model: `texture_mean`, `area_worst`, and `smoothness_worst`. All `_se` features were excluded, so I reran Attempt 1 without the `_se` features. Interestingly, the best combination had the same accuracy score as the randomized approach in Attempt 2 (98.25%) and 2 of the 3 variables from the original report were included (`texture_mean` and `smoothness_worst`).

With this in mind, I replaced radius with area so that all 3 of the original report's features could be included. This resulted in the highest accuracy yet: 99.12%. The combination moving into the final modeling phase would be `area_worst`, `compactness_mean`, `concavity_mean`,

fractal_dimension_mean, smoothness_worst, symmetry_mean, and texture_mean.

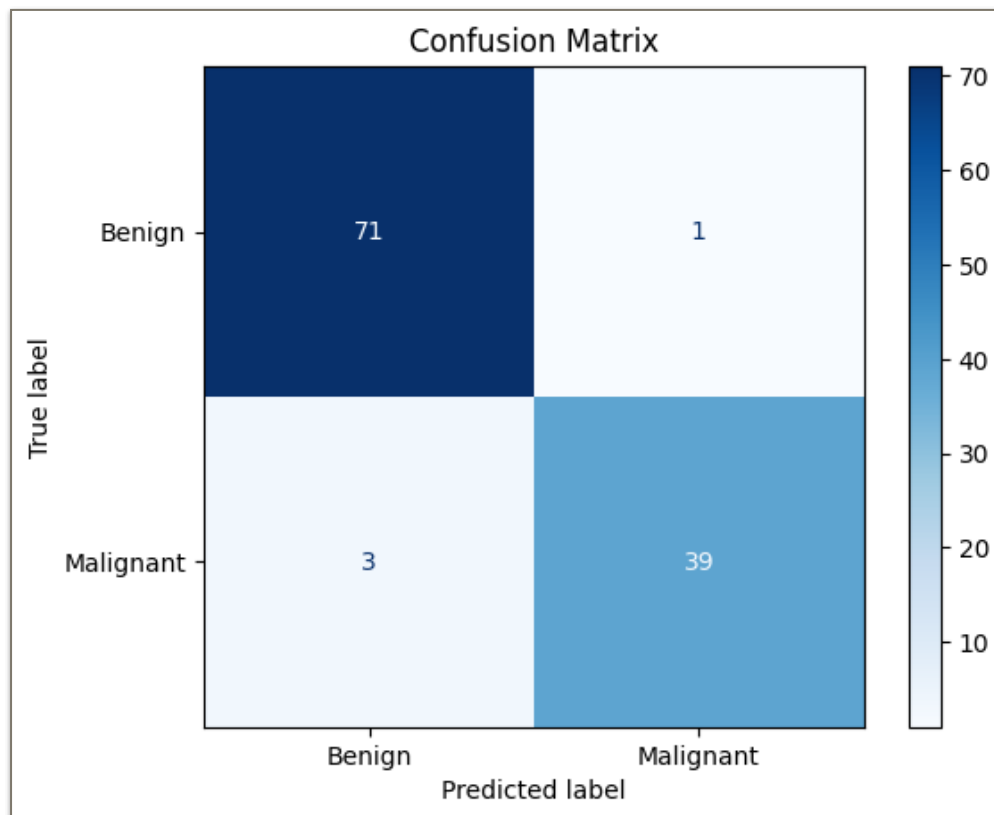
Modeling

The final model required some tuning to handle the imbalanced target variable. Recall that there were 357 benign instances and 212 malignant instances. To address this, I stratified y in the training and testing data and used StratifiedKFold cross validation with 5 splits. I employed GridSearchCV to determine the hyper parameters that would maximize the F1 score.

```
Best Parameters:
{'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 5, 'n_estimators': 100}

Model Evaluation Metrics:
Accuracy: 0.9649
Precision: 0.9750
Recall:    0.9286
F1 Score:  0.9512
ROC AUC:   0.9957
```

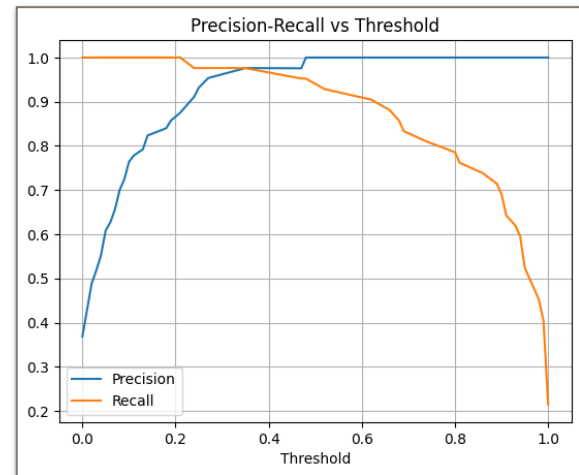
Results of this GridSearchCV



Confusion matrix of the resulting model.

While these results were strong, they did not have sufficient recall, and 3 malignant instances were missed. I employed 2 methods to maximize recall:

1. I first emphasized recall in GridSearchCV instead of the F1 score and assigning class_weight to balanced in the RandomForestClassifier. This eliminated the misclassified benign instance but failed to change the misclassified malignant instances.
2. I manually tuned the threshold to boost recall without sacrificing precision. Using a threshold of 0.2 completely eliminated false negatives but introduced 6 false positives. Given the financial and emotional burdens of false positives, this was not acceptable. However, using a 0.3 resulted in the strongest model yet with only 1 false negative and 1 false positive.



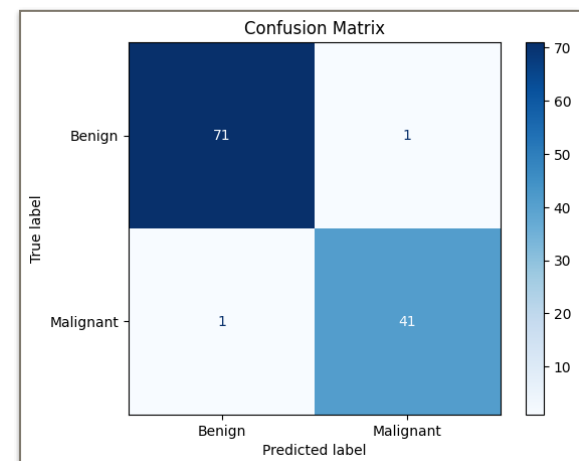
```

Model Evaluation Metrics:
Accuracy: 0.9825
Precision: 0.9762
Recall: 0.9762
F1 Score: 0.9762
ROC AUC: 0.9980

Classification Report:

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	precision	recall	f1-score	support
0	0.99	0.99	0.99	72
1	0.98	0.98	0.98	42
accuracy			0.98	114
macro avg	0.98	0.98	0.98	114
weighted avg	0.98	0.98	0.98	114



Results of the final model. Accuracy: 98.25%. Sensitivity (i.e., precision): 97.62%. Specificity (i.e., recall): 97.62%

Final Model Summary

1. Train test split with 80/20 ratio and stratified target variable.
2. Random forest classifier with n_estimators=100, max_depth=None, min_samples_split=2, min_samples_leaf=1, class_weight='balanced'.
3. Threshold > 0.3
4. For reproducibility, random_state=42 was used throughout this report.

References

1. Street, Nick & Wolberg, William & Mangasarian, O. (1999). Nuclear Feature Extraction For Breast Tumor Diagnosis. Proc. Soc. Photo-Opt. Inst. Eng. 1993. 10.1117/12.148698.
2. Tin Kam Ho, "Random decision forests," Proceedings of 3rd International Conference on Document Analysis and Recognition, Montreal, QC, Canada, 1995, pp. 278-282 vol.1, doi: 10.1109/ICDAR.1995.598994.
3. Wolberg, W., Mangasarian, O., Street, N., & Street, W. (1993). Breast Cancer Wisconsin (Diagnostic) [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5DW2B>.